Final Project Integrity and AI Usage Statement

ITC-C508 (Deep Learning and NLP) · Final Project: RAG Chatbot Development and Evaluation

Group Purple R · College of Computer Studies, Jose Rizal University

Each member completes and signs section C for themselves. A single signature does not cover the group.

A. Authenticity of the Submitted Work

- The work submitted for this project is our own. Where we used material written by others, it is cited.
- Every figure reported in the IEEE paper comes from a run that actually took place. The runs are recorded in PurpleR_Experiment_Logs.xlsx, which carries the timestamp, the hardware, the library versions and the per-stage outputs of the run the paper reports.
- Three measured runs were performed. All three are reported, including the second, whose results were worse than the first. No run was discarded, and no figure was selected from among several runs.
- The evaluation set was written before the system was measured against it, and no question was changed after seeing a result. Two answer keys were corrected before any reported run, for the reasons given in section II of the paper.

B. Disclosure of Generative AI Use

Generative AI tools were used in this project. The table below records where, and is completed individually in section C.

Where AI was used	What it did	What we did
Code	Drafted parts of the notebook, the scoring functions and the build scripts; suggested the dtype selection and the digit-boundary matching rule.	Specified the design, reviewed every line, ran it, found and fixed the defects listed in the MRM, and can explain what each section does.
Writing	Drafted and edited the IEEE report, the data-package documentation and the handover notes.	Supplied the findings and the numbers, checked every figure against the measured output, and corrected claims that overstated the results.
Analysis	Proposed explanations for observed failures, including the diagnosis that retrieval rather than generation is the limiting factor.	Verified each explanation against the recorded data before accepting it; rejected one proposed prompt fix after it measured worse.
Not used for	The evaluation questions and evidence anchors were authored by a group member against the corpus. No reported measurement was produced or estimated by an AI tool; every number comes from an executed run.	

We understand that disclosing the use of these tools is required, and that the responsibility for the accuracy of everything submitted remains ours.

C. Individual Declaration

Each member completes one block. Describe your own use of generative AI in your own words, in the space provided, then sign.

Leader	Mark Emmanuel P. Sagarino
Component I own and can defend	The retrieval-augmented pipeline and the evaluation harness. This covers document loading and splitting, embedding, the Chroma index and its stored-count assertion, the retrieval and generation function, the five-configuration chunking experiment and the selection of the operating point, the three prompt versions and their measurement, and the scoring design — verbatim evidence anchors as ground truth, the answer-key semantics, the treatment of unanswerable questions as undefined rather than zero, and refusal measured in both directions. I also wrote the analysis and this report, and integrated the components contributed by Leliza and Dizon.
My use of generative AI	I used Claude extensively throughout this project, and the work would look different without it. It drafted most of the notebook code, the scoring functions and the build scripts, and it wrote the first and subsequent drafts of the IEEE report, the data-package documentation, the Member Responsibility Matrix and this statement. It proposed the compute-capability check for selecting float16 over bfloat16, the word-boundary rule for digit-only answer keys, and the diagnosis that retrieval rather than generation is the limiting factor in our results. What I did was specify the domain and the research question, decide the experimental design, review the code and the output at each stage, direct the three prompt iterations, and judge which proposals to accept. I rejected one prompt revision after it measured worse than the version it replaced, and I decided to stop after the third run and report the failed second one rather than present only the best result. No reported number was produced, estimated or edited by an AI tool. Every figure in the paper comes from a run that executed on Google Colab, and the timestamps, hardware and library versions of that run are recorded in PurpleR_Experiment_Logs.xlsx. The twelve evaluation questions and the twenty-one evidence anchors were written by Leliza against the corpus, not generated. I understand that the responsibility for the accuracy of everything submitted under my name is mine, and I am prepared to explain and defend the component described above.
I declare that the above is accurate and complete.	Confirmed — Mark Emmanuel P. Sagarino

Signature	
Date	11/08/2026

Member	Ram Kelly Jerzen Leliza
Component I own and can defend	Deliverable 5, the evaluation test set. Twelve questions — three easy, three average, three difficult and three unanswerable — each with a reference answer, answer keys and verbatim evidence anchors drawn from the corpus, twenty-one anchors in total, together with the written rationale for every question and the verification output. I can explain why each question is in the set, why the difficult questions turn on exclusions rather than on wording, and why two anchors that look interchangeable are not: the phrase "a proportionate price reduction" appears in both Article 100 and Article 102 of RA 7394, so I replaced it with a neighbouring phrase whose comma makes it unique to Article 100. That constraint — every anchor must occur exactly once in the corpus, and none may cross a line break — is mine, and it corrected an anchor that would otherwise have inflated the relevance counts.
My use of generative AI	I used Claude while building the test set. It helped me phrase the questions in a buyer's voice and draft the rationale document. What I did myself was read the five source documents, choose which provisions each question would turn on, copy the anchors verbatim from the files, and run verify_anchors.py until all twenty-one passed. The questions and the answer keys are my decisions about what a correct answer must contain.
I declare that the above is accurate and complete.	Yes — Ram Kelly Jerzen Leliza
Signature	
Date	11/08/2026

Member	Sean John C. Dizon

Component I own and can defend	Deliverable 4 — Experiment Logs and Checkpoints. The nine-sheet workbook PurpleR_Experiment_Logs.xlsx written by Section 15 of the notebook, the nine stage screenshots, and the run notes. I ran the notebook on Google Colab (Tesla T4) — three runs, the third of which the paper reports. I can explain what each sheet records, why each check in the Section 4 pre-flight exists, and how the evaluation columns are scored, including why answer quality is left blank rather than zero for the three unanswerable questions.
My use of generative AI	I used Claude (Anthropic) throughout my component, for preparation and verification rather than for producing the notebook. Reviewing my deliverable README against the data package, which surfaced three inconsistencies corrected before the run: the question count (twelve against ten), the notebook filename, and whether a CPU run was required. Verifying all 21 evidence anchors against the corpus — each occurring exactly once, and none crossing a line break. Reviewing the group's answer-scoring function, which surfaced four defects corrected before the run: a crash on the unanswerable rows, two answer keys that the reference answers themselves could not satisfy, and a digit variant that matched inside statute numbers. Writing purpler_testset_check.py, the pre-flight script included in the data package and run as Section 4 of the notebook. I reviewed it and ran it; I did not write it line by line. Drafting some of my written updates to the group, and drafting this declaration. Claude did not run the notebook, take the screenshots, write the notebook, or author the test set, and supplied none of the figures in my run notes.
I declare that the above is accurate and complete.	Yes — Sean Dizon
Signature	
Date	11/08/2026